to-locate boundary is somehow calling the shots regarding what happens here in the bulk. Does the holographic principle imply that *that* sense of control and autonomy is illusory? Or is it better to think of holography as articulating a kind of duality in which, on the basis of taste—not of physics— one can choose a familiar description in which the fundamental laws operate here in the bulk (which aligns with intuition and perception) or an unfamiliar description in which fundamental physics takes place on some kind of boundary of the universe, with each viewpoint being equally valid? These are essential questions that remain controversial.

But in 1997, building on earlier insights of a number of string theorists, the Argentinian physicist Juan Maldacena had a breakthrough that dramatically advanced thinking on these matters. His discovery is not directly relevant to the question of holography's role in our real universe, but in the time-honored fashion of physics, he found a hypothetical context—a hypothetical universe—in which abstract musings on holography could be made both concrete and precise using mathematics. For technical reasons, Maldacena studied a hypothetical universe with four large space dimensions and one time dimension that have uniform negative curvature—a higher dimensional version of the Pringle's potato chip, Figure 8.6c. Standard mathematical analysis reveals that this fivedimensional spacetime has a boundary[7] that, like all boundaries, has one dimension less than the shape it bounds: three space dimensions and one time dimension. (As always, higher-dimensional spaces are hard to envision, so if you want a mental picture, think of a can of tomato soup—the three-dimensional liquid soup is analogous to the five-dimensional spacetime, while the two-dimensional surface of the can is analogous to the four-dimensional spacetime boundary.) After including additional curled-up dimensions as required by string theory, Maldacena convincingly argued that the physics witnessed by an observer living within this universe (an observer in the "soup") could be completely described in terms of physics taking place on the universe's boundary (physics on the surface of the can).

Although it is not realistic, this work provided the first concrete and mathematically tractable example in which the holographic principle was explicitly realized.[8] In doing so, it shed much light on the notion of holography as applied to an entire universe. For instance, in Maldacena's